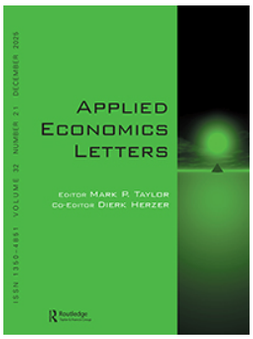

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Geopolitical risk and long-term stock market volatility: evidence from the Korean Peninsula using a GJR-GARCH-MIDAS framework

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ABSTRACT

This study uses the GJR-GARCH-MIDAS model to empirically analyse the impact of Geopolitical Risk from North Korea (GPRNK) on the long-term volatility of South Korea's KOSPI stock market. The estimation results based on daily returns and monthly GPRNK indicators indicate that realized volatility (RV) has a significant positive impact on long-term volatility. In contrast, GPRNK has a significant negative effect on long-term volatility. This result suggests that, in the context of frequent exposure to geopolitical risk (GPR), investors tend to adopt more conservative trading strategies, thereby reducing market volatility. Further analysis reveals that incorporating GPRNK into the model significantly enhances the predictive performance of long-term volatility, with a maximum predicted loss reduction of 4.2%. In summary, GPRNK has played a significant role in shaping the volatility characteristics of the South Korean stock market, providing valuable empirical evidence for improving long-term volatility forecasts.

KEYWORDS

Geopolitical risk from North Korea (GPRNK); KOSPI stock market; long-term volatility; GJR-GARCH-MIDAS model

JEL CLASSIFICATION

C58; G15; G17; F51; F37

I. Introduction

Geopolitical risk (GPR) has long been regarded as a significant exogenous shock factor affecting global financial markets (Salisu et al. 2022). As an economy highly dependent on foreign trade, South Korea's financial markets are susceptible to external political events. Due to the long-standing military standoff and strategic uncertainty on the Korean Peninsula, geopolitical events such as North Korea's nuclear tests (Huh and Pyun 2018), missile launches (Lee 2024), and border conflicts (Son and Moon 2021) are often viewed by the market as low-probability but high-impact 'tail risks', which frequently triggers risk-averse sentiment among investors. This sentiment, through mechanisms such as adjustments in risk preferences and expectations, gradually translates into a reassessment of long-term risks (Salisu, Gupta, and Bouri 2023) .[1]

This study investigates whether the escalation of geopolitical tensions between North and South Korea has a significant impact on the long-term volatility of the South Korean KOSPI stock market. Previous studies indicate that heightened tensions between North and South Korea often trigger short-term market panic and increased volatility (Pyo 2021). However, the mechanisms of manifestation and transmission of such shocks on long-term volatility remain inadequately explained. As an exogenous, non-economic shock variable, GPR can significantly enhance the predictive accuracy of volatility models in high-volatility environments, thereby supplementing the explanatory power of traditional macroeconomic indicators (Foglia et al. 2025; Li and Liu 2025). This provides a more robust theoretical foundation and behavioural mechanism explanation for the present study. In addition, existing studies generally overlook the estimation bias that may arise from the mismatch between macro variables and the frequency of high-frequency financial data (Galvão 2013).[2]

In response to this research gap, this study adopts the GJR-GARCH-MIDAS model framework (Conrad and Kleen 2020). It combines daily KOSPI return volatility with the monthly Geopolitical Risk from North Korea (GPRNK) index to systematically examine the dynamic predictive role of GPRNK on long-term volatility.[3] This model captures high-frequency market information in short-term volatility

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[1]A detailed account of the geopolitical background is provided in *Appendix A*, available in the online supplementary material.
[2]A comprehensive literature review is provided in *Appendix B*, available in the online supplementary material.
[3]Data sources and descriptive statistics are provided in *Appendix C*, available in the online supplementary material.

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modelling while incorporating low-frequency GPR indicators into long-term components. It provides a practical approach for handling data across different frequencies, thereby revealing the long-term impact mechanisms of GPR while preserving high-frequency volatility information.

The results reveal that realized volatility (RV) significantly contributes to long-term volatility, while GPRNK exerts a significant negative effect. This suggests a convergence of market volatility amid persistent geopolitical tensions, likely driven by investors' adaptive behaviours and reduced sensitivity to external shocks (Pyo 2021). Moreover, incorporating GPRNK into the two-factor model enhances forecast accuracy, reducing the maximum forecast loss by approximately 4.2% relative to the benchmark. These findings underscore the importance of incorporating geopolitical indicators into long-term volatility modelling.

This study contributes to the literature on GPR and financial market volatility in three ways. First, it applies the GJR-GARCH-MIDAS model to the Korean Peninsula, integrating high-frequency market data with low-frequency geopolitical shocks and addressing mixed-frequency modelling gaps. Second, results show that GPRNK significantly dampens long-term volatility, reflecting adaptive market behaviour and investor habituation to risk. Third, incorporating GPRNK improves long-term volatility forecasts, underscoring its value for financial prediction, risk management, and stakeholder decision-making.

II. Methodology

In-sample prediction test

This study draws on the methods of Engle, Ghysels, and Sohn (2013) and Dai et al. (2022) to construct two types of models, namely fixed window and rolling window models, to more comprehensively assess the dynamic impact of exogenous variables on stock market volatility.[4]

Fixed window model

Firstly, this study constructs single-variable and double-variable GJR-GARCH-MIDAS models to analyse the impact of GPR on the daily return volatility $(r_{i,t} = (lnP_t - lnP_{t-1}) * 100)$ of the KOSPI index.

$$r_{i,t} = \mu + \sqrt{\tau_t * g_{i,t}}\varepsilon_{i,t} \quad \forall \mathrm{i} = 1, \ldots, \mathrm{N_t} \tag{1}$$

Among these, the short-term volatility term $g_{i,t}$ adopts the GJR-GARCH (1,1) structure (Conrad and Kleen 2020):

$$g_{i,t} = \left(1 - \alpha - \frac{\gamma}{2} - \beta\right) + \left(\alpha + \gamma * I_{\left\{\varepsilon_{i-1,t} < 0\right\}}\right)\frac{\varepsilon^2_{i-1,t}}{\tau_t} + \beta g_{i-1,t} \tag{2}$$

The long-term volatility term τ_t is modelled using MIDAS regression, introducing monthly exogenous variables like RV and GPRNK:

$$ln\tau_t = m + \theta \sum_{k=1}^{K} \varphi_k(w_1, w_2) X_{t-k} \tag{3}$$

In the weighting function, set to ensure that the influence of each exogenous variable on long-term volatility shows a monotonic downward trend. The simplified weighting function can be expressed as:

$$\varphi_k(w_2) = \frac{\left(1 - \frac{k}{K}\right)^{w_2 - 1}}{\sum_{j=1}^{K}\left(1 - \frac{j}{K}\right)^{w_2 - 1}} \tag{4}$$

Furthermore, this study introduces two types of variables, RV $(RV = \sum_{i=1}^{N_t} r^2_{i,t})$ and GPRNK $(\Delta\mathrm{GPRNK} = (\mathrm{lnGPRNK_t} - \mathrm{lnGPRNK_{t-1}}) * 100)$, into the long-term volatility part to construct a two-factor GJR-GARCH-MIDAS model:

$$ln\tau_t = m + \theta_1 \sum_{k=1}^{K} \varphi_{1k}(w_{1,1}, w_{1,2}) RV_t + \theta_2 \sum_{k=1}^{K} \varphi_{2k}(w_{2,1}, w_{2,2}) GPRNK_{1,t-k} \tag{5}$$

Therefore, Equations (1), (2), (3), and (4) constitute the single-factor GJR-GARCH-MIDAS model with a fixed window; Equations (1), (2), (5), and (4) constitute the two-factor GJR-GARCH-MIDAS model with a fixed window.

Rolling window model

To enhance model adaptability, this study introduces a rolling window version that only changes

[4]A detailed description of the in-sample prediction test design is provided in *Appendix D* of the online supplementary material.

the long-term volatility regression. This version calculates RV and GPRNK by calculating the moving average of the past 22 trading days ($N' = 22$), enabling the model to respond dynamically to changes in market structure. The specific calculations are as follows:

$$RV_t^{(rw)} = \sum_{j=1}^{N'} r_{i-j}^2; GPRNK_i^{(rw)} = \frac{1}{N'} \sum_{j=1}^{N'} GPRNK_{i-j} \tag{6}$$

The regression form corresponding to long-term volatility is:

$$ln\tau_t^{(rw)} = m^{(rw)} + \theta^{(rw)} \sum_{k=1}^{K} \varphi_k(w_2) X_{t-k}^{(rw)} \tag{7}$$

For the two-factor GJR-GARCH-MIDAS model rolling window version, the expression for long-term volatility $\tau_t^{(rw)}$ is:

$$ln\tau_t^{(rw)} = m^{(rw)} + \theta_1^{(rw)} \sum_{k=1}^{K} \varphi_{1k}(w_{1,2}) RV_t^{(rw)} + \theta_2^{(rw)} \sum_{k=1}^{K} \varphi_{2k}(w_{2,2}) GPRNK_{1,t-k}^{(rw)} \tag{8}$$

Therefore, Equations (1), (2), (7), and (4) constitute the rolling window version of the single-factor GJR-GARCH-MIDAS model; Equations (1), (2), (8), and (4) constitute the rolling window version of the two-factor GJR-GARCH-MIDAS model.

Out-of-sample prediction test

To evaluate out-of-sample forecasting performance, this study uses two complementary methods: the Model Confidence Set (MCS) and the Diebold-Mariano (DM) test. The MCS identifies models with statistically similar predictive accuracy based on selected loss functions.[5] To ensure robustness, the evaluation considers mean squared error (MSE), mean absolute error (MAE), and quasi-likelihood error (QLIKE). The DM test compares model pairs, with significant p-values indicating superior forecasting performance.

III. Empirical results

In-sample estimation results analysis

The results of the single-factor and two-factor GJR-GARCH-MIDAS models, under both fixed-window and rolling-window versions, are reported in Table 1.

In the short-term component, the asymmetry term γ is significantly positive, confirming the existence of a typical leverage effect in the Korean stock market, i.e. negative shocks have a more substantial impact on volatility than positive shocks. Additionally, the parameter estimates $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta + \gamma/2 < 1$ (but close to 1), indicating that short-term volatility is highly persistent and exhibits a pronounced 'volatility clustering' characteristic.

In the long-term component, the θ coefficient for RV is significantly positive, indicating that short-term fluctuations will be transformed into long-term market risk through the cumulative effect of information, which supports the risk premium theory (Bordalo et al. 2020). On the contrary, the θ coefficient of ΔGPRNK is significantly negative, indicating that geopolitical tensions do not amplify long-term market volatility, which may reflect investors' adaptive expectations and risk-averse behaviour. This aligns with the 'adaptive market hypothesis', which posits that market participants adjust their behaviour based on experience and learning (Rönkkö et al. 2024).

The two-factor model further validates the above relationship: the positive impact of RV on long-term volatility remains significant. At the same time, the negative effect of ΔGPRNK persists even after controlling for RV, indicating that the two variables have independent explanatory power for long-term volatility, which highlights the necessity of considering both endogenous information from financial markets and exogenous macroeconomic or geopolitical information in the analysis.

In terms of model fitting indicators, the rolling window version outperforms the fixed window model in terms of log-likelihood (LLH), AIC, and BIC, indicating its superiority in dynamically tracking market structural changes and enhancing model adaptability. Therefore, in the subsequent out-of-sample forecasting analysis, this study

[5]A detailed description of the out-of-sample prediction test design is provided in *Appendix E*, available in the online supplementary material.

Table 1. In-sample estimation results for the model.

	Model 1 RV	Model 2 RV^{rw}	Model 3 ΔGPRNK	Model 4 ΔGPRNK^{rw}	Model 5 $RV + \Delta$GPRNK	Model 6 $(RV + \Delta\text{GPRNK})^{rw}$
μ	0.029**	0.029***	0.026*	0.028*	0.029***	0.030***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
α	0.008	0.008	0.035***	0.035***	0.016***	0.005
	(0.007)	(0.007)	(0.007)	(0.007)	(0.006)	(0.007)
β	0.834***	0.835***	0.895***	0.894***	0.864***	0.838***
	(0.011)	(0.011)	(0.007)	(0.007)	(0.009)	(0.012)
γ	0.179***	0.178***	0.107***	0.106***	0.144***	0.176***
	(0.014)	(0.014)	(0.011)	(0.010)	(0.012)	(0.014)
m	−0.436***	−0.454***	0.380***	0.312***	−0.485***	−0.490***
	(0.065)	(0.065)	(0.111)	(0.110)	(0.078)	(0.065)
θ_{RV}	0.016***	0.016***			0.017***	0.016***
	(0.001)	(0.001)			(0.001)	(0.001)
θ_{GPRNK}			−0.075***	−0.116***	−0.122**	−0.105***
			(0.026)	(0.031)	(0.022)	(0.022)
$w_{2,\text{RV}}$	4.531***	4.836***			1.000***	3.907***
	(0.628)	(0.631)			(0.178)	(0.514)
$w_{2,\text{GPRNK}}$			1.000***	1.000***	1.000***	1.000***
			(0.167)	(0.126)	(0.092)	(0.100)
LLH	−6776.137	−6722.552	−6803.259	−6750.022	−6775.112	−6714.583
AIC	13566.274	13459.104	13620.518	13514.044	13568.224	13447.166
BIC	13611.369	13504.160	13665.612	13559.101	13626.203	13505.096

Note: This table reports parameter estimates for one- and two-factor models. LLH is the log-likelihood function, AIC denotes the Akaike Information Criterion, and BIC denotes the Bayesian Information Criterion. Numbers in parentheses are standard errors. ***, **,and * denote 1%, 5%, and 10% significance levels, respectively. No superscripts indicate fixed window parameters, while superscripts with the superscript "rw" indicate rolling window parameters. Monthly values in the rolling window version will be copied to the values for each day of the corresponding month.

primarily focuses on the three model configurations under the rolling window version.[6]

Out-of-sample forecast analysis

In model evaluation, out-of-sample prediction ability is the core standard for measuring model performance, directly reflecting the accuracy of prediction results and the effectiveness of model settings. To this end, this study employs the rolling window method to generate daily progressive predictions for the last 5 years (1,165 trading days) of the entire sample data, systematically testing the predictive performance of the model in an actual application.

Table 2 reports the MCS test results. Under the MSE loss function, all models passed the significance test, among which Model 6 (simultaneously introducing RV and GPRNK) performed best with the smallest prediction error, indicating that it has a strong advantage in capturing market volatility. Under the MAE standard, Model 2 (containing only RV) had the best prediction performance, highlighting the importance of RV for long-term volatility prediction. Under the QLIKE loss function, all models passed the MCS test, meaning all MCS p-values exceeded 0.10, with Model 6 demonstrating the best predictive performance.

Table 2. MCS test results.

	MSE		MAE		QLIKE	
MCS-1165 days	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
GJR-GARCH(1,1)	**0.353**	**0.205**	0.041	**0.132**	**0.175**	**0.115**
Model 2	**0.672**	**0.672**	**1.000**	**1.000**	**0.825**	**0.825**
Model 4	**0.353**	**0.205**	0.011	0.009	**0.189**	**0.126**
Model 6	**1.000**	**1.000**	**0.591**	**0.591**	**1.000**	**1.000**

Note: The numbers in the table are p-values. Numbers greater than 0.10 are in bold and underlined, indicating that the corresponding model performs significantly better in the model set. The time horizon is 1165 days.

Additionally, this study employs the GJR-GARCH (1,1) model as the benchmark and uses the Diebold-Mariano (DM) test to compare prediction errors further. Table 3 shows that under the MSE and QLIKE metrics, Model 6 reduces prediction errors by approximately 4.2% and 2.9%, respectively, significantly outperforming the benchmark model. This result validates the effectiveness of incorporating GPR as a key driver of long-term volatility and jointly modelling it with high-frequency market information (RV).

[6]The relationship between long-term volatility and total volatility for various models, the MIDAS regression lag weight distribution chart, and related explanations are provided in *Appendix F*, available in the online supplementary material.

Table 3. DM test results.

DM-1165 days	MSE	MAE	QLIKE
Model 2	0.961**	0.969***	0.974**
	(−2.489)	(−6.098)	(−2.424)
Model 4	0.994	1.005	0.994
	(−1.387)	(1.855)	(−0.987)
Model 6	0.958***	0.974***	0.971**
	(−3.323)	(−3.834)	(−2.083)

Note: This table shows the results of the Diebold-Mariano forecast evaluation relative to various models for which the GJR-GARCH(1,1) model is used as a benchmark. A ratio of less than 1 indicates that the model predicts volatility better than the benchmark model.DM test statistics are shown in parentheses. The time horizon is 1165 days. ***,**,* denote rejection of the original hypothesis at the 1%, 5% and 10% significance levels, respectively.

To sum up, compared with relying solely on RV, integrating low-frequency exogenous variables such as GPRNK significantly enhances the ability to characterize complex market dynamics and improves prediction accuracy and model robustness.[7]

IV. Conclusions

This study employs the GJR-GARCH-MIDAS model to investigate the impact of GPRNK on the long-term volatility of the South Korean stock market. Results show that short-term RV feeds into long-term volatility through accumulated market information, while GPRNK has a significant negative effect, reflecting investor adaptation in the face of persistent geopolitical tensions. Combining both factors improves predictive accuracy by about 4.2%, highlighting the value of integrating high-frequency financial data with low-frequency geopolitical indicators.

The findings provide practical implications for policymakers, regulators, risk managers, investors, and international stakeholders. Authorities should monitor geopolitical risks, establish contingency mechanisms, and stabilize market expectations, while risk managers and firms incorporate these risks into hedging, stress testing, and strategic planning. These coordinated efforts enhance market resilience. Limitations remain, as the study focuses only on the Korean Peninsula, may not capture nonlinear or spillover effects, and relies on the limited GPRNK index. Future work could adopt broader models and expand cross-market analysis to further enhance understanding.[8]

Author contributions

CRediT: **Shun Li:** Data curation, Methodology, Writing – original draft, Writing – review & editing; **Yang Liu:** Supervision, Validation, Writing – original draft, Writing – review & editing.

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Availability of data and materials

The data presented in this study are available on request from the corresponding author.

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[7]A detailed description and analysis of the robustness checks is provided in *Appendix G*, available in the online supplementary material.
[8]A comprehensive discussion of findings and their implications is provided in *Appendix H*, available in the online supplementary material.

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